

Wi-Fi Security Analysis Report

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Property	Value
SSID	P420-2
BSSID	4C:12:E8:E8:21:38
Security Standard	High Risk (Mixed WPA1/WPA2 AES)
Signal Strength	-52.00 dBm
Band / Channel	2.4GHz / channel 4
Security Score	35/100

Positive Security Findings

[+] Strong AES/CCMP encryption: The current industry standard for data privacy.

Risks & Recommendations

- [!] **RISK:** Legacy Mixed Mode: Supports outdated WPA1 protocols.
Suggestion: Update router configuration to 'WPA2-AES Only'. Supporting WPA1 makes your network vulnerable to older exploits.
- [!] **RISK:** Personal PSK: Vulnerable to offline dictionary/brute-force attacks.
Suggestion: Use a complex password (12+ characters, mixed cases, numbers, symbols) and change it periodically.
- [!] **RISK:** Missing MFP: Vulnerable to de-authentication (disconnection) attacks.
Suggestion: Enable 'Management Frame Protection' (802.11w) in security settings to prevent attackers from forcibly disconnecting your devices.

Full Technical Details (Raw Metadata)

The following data is extracted directly from the Wi-Fi beacon/probe response frames:

Country:

Code: US

Environment: Indoor/Outdoor

constraints:

- Channels [1 - 11] @ 30 dBm

DS Parameter set: channel 4

ERP: Barker_Preamble_Mode

Extended capabilities:

flags:

- HT Information Exchange Supported
- BSS Transition

Extended supported rates: 6.0 12.0 24.0 48.0

HT capabilities:

Capabilities:

Max AMSDU length: 3839 bytes

flags:

- RX LDPC
- HT20
- SM Power Save disabled
- RX HT20 SGI
- TX STBC
- RX STBC 1-stream
- No DSSS/CCK HT40

info: 0x1ad

HT RX MCS rate indexes supported: 0-15

Maximum RX AMPDU length 65535 bytes (exponent: 0x003)

Minimum RX AMPDU time spacing: 4 usec (0x05)

flags:

- HT TX MCS rate indexes are undefined

HT operation:

HT protection: no

L-SIG TXOP Prot: 0

OBSS non-GF present: 0

PCO active: 0

PCO phase: 0

RIFS: 0

STA channel width: 20 MHz

STBC beacon: 0

dual CTS protection: 0

dual beacon: 0

non-GF present: 1

primary channel: 4

secondary channel offset: no secondary

Information elements from Probe Response frame:

RSN:

Authentication suites: PSK

Capabilities: 1-PTKSA-RC 1-GTKSA-RC (0x0000)

Group cipher: CCMP

Pairwise ciphers: CCMP

Version: 1

SSID: P420-2

Supported rates: 1.0* 2.0* 5.5* 11.0* 9.0 18.0 36.0 54.0

TSF: 1277940196190 usec (14d, 18:59:00)

WMM:

BE: CW 15-1023, AIFSN 3

BK: CW 15-1023, AIFSN 7

VI: CW 7-15, AIFSN 2, TXOP 3008 usec

VO: CW 3-7, AIFSN 2, TXOP 1504 usec

info: Parameter version 1

WPA:

Authentication suites: PSK

Group cipher: CCMP

Pairwise ciphers: CCMP

Version: 1

beacon interval: 100 TUs

capability: ESS Privacy ShortSlotTime (0x0411)

freq: 2427.0

last seen:

- 55413.220s [boottime]
- 2177 ms ago

signal: -52.00 dBm